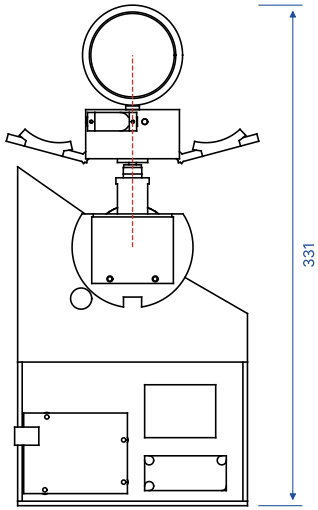
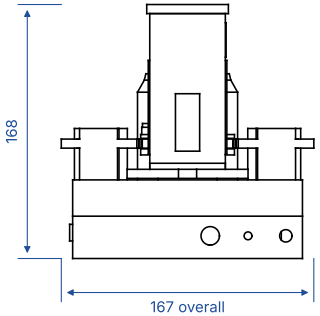
CA-000 Concept A assembly, HOME pose (arm at START)

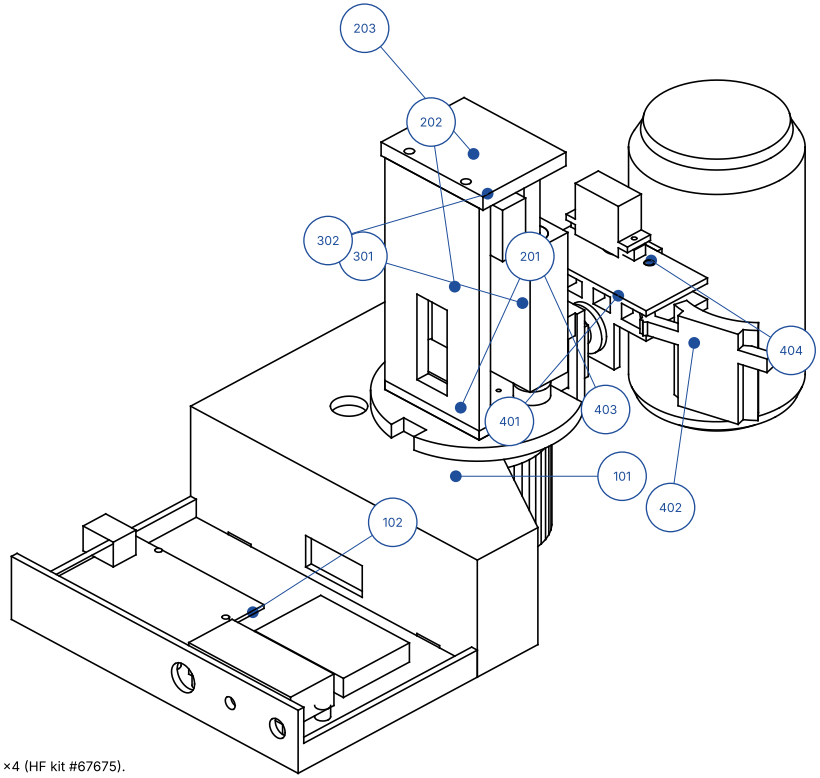
TOP VIEW



R 127 mm (5.0 in): turntable axis → can centre (dashed)



FRONT VIEW



ISOMETRIC (not to scale)

PARTS LIST (full BOM: docs/05_bom.md)

CA-101	Base	PETG	1
CA-102	Electronics tray	PETG	1
CA-201	Turret plate	PETG	1
CA-202	Spine	PETG	1
CA-203	Top cap	PETG	1
CA-301	Carriage + arm	PETG	1
CA-302	Lift link	PETG	1
CA-401	Gripper frame	PETG	1
CA-402	Finger, right	PETG	1
CA-403	Finger, left	PETG	1
CA-404	Can bumper	silicone	1
CA-P02	Servo MG996R (turntable)	buy	1
CA-P03	Servo MG90S (lift, wrist)	buy	2
CA-P04	Servo SG90 (gripper)	buy	1
CA-P05	Bearing 6807-2RS	buy	1
CA-P06	Linear bearing LM6UU	buy	4
CA-P07	Steel rod 6 × 112 mm	buy	2
CA-P01	Arduino Uno R3	buy	1
CA-P09	Buck converter XL4015	buy	1

Balloon numbers = part number without the CA- prefix. Fasteners: #6-32 × ½″ ×8, #8-32 × ½″ ×4 (HF kit #67675).
Motion ranges: turret 0 → −90°, lift 0 → 15.3 mm, wrist 0 → 180°, fingers 0 → 75°.

PART NO.	**CA-000**
TITLE	**Concept A assembly (HOME pose)**
MASS	≈ 571 g robot + 15 g can (CAD)
SCALE	1:5 (front, top)
UNITS	mm
SOURCE	cad/out/concept_a_detail_assembly.step
PROJECTION	Third angle ⊕◁
SHEET	1 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02